



DOW™ Electrical & Telecommunications HFDB-4201 EC Crosslinkable Power Cable Insulation Compound

Overview

HFDB-4201 is a long-life, unfilled, crosslinkable, low density, polyethylene insulation compound designed for medium and high voltage (5 kV and higher) power cable insulation applications. It has a very low level of contamination and bears the designation Extra Clean (EC). HFDB-4201 EC has been designed to have an extremely low level of additive bloom for a long storage life, low dusting, and an enhanced degree of scorch retardance for fine mesh filtering and long production run lengths during cable manufacture.

Specifications

HFDB-4201 EC is designed for use in power distribution and sub-transmission cables. Cables insulated with HFDB-4201 EC, using sound commercial fabrication manufacturing practice, would be expected to meet the latest editions of the following specifications and regulations:

- ANSI/ICEA: S-94-649, S-97-682, S-93-639 / NEMA WC74
- AEIC: CS8
- RUS: 50-70 (U-I)
- CEA: WCWG-01, WCWG-02
- UL 1072
- IEC: 60502, 60840
- CENELEC: HD 620 S1/A3, HD 632 S2, HD 629 S1
- DIN VDE: 0207, DIN VDE 0276-620
- BSI BS: 6622
- GB/T 12706

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density	0.920 g/cm ³	0.920 g/cm ³	ASTM D792
Degree of Crosslinking - Extractables	< 20 %	< 20 %	ASTM D2765A
Mechanical	Nominal Value (English)	Nominal Value (SI)	Test Method
Tensile Strength	3000 psi	20.7 MPa	ASTM D638
Tensile Elongation (Break)	530 %	530 %	ASTM D638
Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Hot Creep (392°F (200°C))	< 100 %	< 100 %	ICEA T-28-562
Hot Set (392°F (200°C))	< 10 %	< 10 %	ICEA T-28-562
Electrical	Nominal Value (English)	Nominal Value (SI)	Test Method
Volume Resistivity (73°F (23°C))	> 1.0E+15 ohms·cm	> 1.0E+15 ohms·cm	ASTM D257
Dielectric Strength			ASTM D149
0.125 in (3.18 mm) ¹	660 V/mil	26 kV/mm	
0.125 in (3.18 mm) ²	590 V/mil	23 kV/mm	
Dielectric Constant (60 Hz)	2.30	2.30	ASTM D150
Dissipation Factor (60 Hz)	3.0E-4	3.0E-4	ASTM D150

Additional Information

Nominal property values above represent tests on molded stress-relieved slabs. Cure times were 15 minutes at (175°C).

Extra-Clean Requirements

- HFDB-4201 EC meets the strictest standards for cleanliness established for an unfilled, crosslinkable cable insulation compound. Throughout the production process, the product is tested to ensure the highest level of cleanliness. Extruded tapes are scanned by an automatic inspection system in a class 10,000 clean room. The purity data is managed using an acceptance sampling plan which ensures that the product in the shipping container meets or exceeds extra-clean standards.
- Maximum Allowable in 1.0 Kg Tape: HFDB-4201 EC
 - 125-250 µm: 2
 - 250-625 µm: 0
 - >625 µm: 0
- In addition, as HFDB-4201 EC is packaged, a continuous side stream of pellets is analyzed using an electronic pellet inspector. A review of detected contamination is incorporated into our EC quality program.

Storage

- The environment or conditions of storage greatly influences the recommended storage time. Storage under extreme conditions may affect the quality, processing, or performance of the product. Storage should be in accordance with good manufacturing practices. The recommended storage conditions are dry conditions with temperatures between 50°F and 86°F (10°C and 30°C). When stored under these conditions, the product may be used by the customer for up to one year from the date of sale or two years from the date of manufacture, whichever comes first. It is recommended that the practice of using the product on a first-in / first-out basis be established.

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

¹ Method A (Short-Time)

² Method B (Step-by-Step)

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